

$$\begin{aligned}
 M_{t_1, \text{qs}} &= K_{R, t_1} t_1^\top (e_R - e_{R, 0}) \\
 &= (15 \text{ N m/rad}) \cdot \left(6.57 \cdot \frac{\pi}{180} \text{ rad} \right) \approx 1.72 \text{ N m}
 \end{aligned}$$